The Non-AI Baseline: Slow and Unreliable

MRI-ReportGenerator (Cervical Spine MRI Analysis)

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Claim. The non-AI baseline our system augments is *manual radiologist measurement* of cervical spine MRI — a radiologist visually inspects the sagittal T2 series and hand-measures vertebral, disc, canal, and cord geometry. It is the current standard of care, and it is both **slow** and **unreliable**, with inter-observer agreement collapsing for precisely the structures we automate. Below we quantify both costs from primary sources (PMIDs given; authors verified against their PMID). Citations carry honest provenance flags: *[cervical MRI]* where directly on point, and *[borrow]* where the only available figure is from radiograph/CT or the lumbar spine.

1 Cost 1 — Time (slow)

- **Reading time.** Cervical MRI reporting takes ≈ **2.7 min median / 3.8 min mean** of PACS interaction (excluding dictation) — Forsberg et al., 2017 (PMID 27714473).
- **Measurement time. No formal time-motion study** isolates the hand-geometry measurement time for cervical MRI (a documented negative); the only figure is an informal **5–10 min/case** self-report — Zhu et al., 2024 (PMID 38269650, discussion). The closest *formal* comparator is full-spine radiograph alignment measurement at 147 ± 19 s (expert) rising to 232 ± 20 s (resident) — Pucciarelli et al., 2025 (PMID 41003360) *[borrow]*.

A deterministic pipeline computes the same geometry in ≈seconds per case (measured per-component latency; see the deployment/tradeoffs docs), removing the per-case minutes spent on repetitive caliper placement.

2 Cost 2 — Inter-observer variability (unreliable)

The decisive cost is reproducibility. Where two radiologists (or the same radiologist on different days) measure the same study, they disagree — and the disagreement is *worst* for the cord, compression, and disc-degeneration readings that drive myelopathy/radiculopathy decisions. A deterministic pipeline has inter-observer ICC = 1.0 by construction.

Lead with where agreement collapses: cord AP axial ICC **0.66**, cord compression ICC **0.35–0.56**, Pfirrmann-on-cervical κ **0.265**, and Cobb dropping from ∼0.96 (expert pairs) to ∼**0.55** across mixed readers. These are not edge cases — they are the measurements that decide whether a cord is compressed and a disc degenerated.

Measurement (cervical MRI unless flagged)	**Human inter-observer agreement**	**Source (PMID)**
Cord AP diameter	ICC **0.66** (axial) / 0.82 (mid-sag)	Grochmal 2018 (29913296)
Cord compression (trauma)	ICC **0.35–0.56**	Fehlings 2006 (16816769)
Cord compression ratio / MSCC	ICC 0.79–0.86	Karpova 2013 (22772577); 2010 (23637669)
Disc-degeneration grade (κ)	**0.265** (Pfirrmann-on-cervical); 0.364 (Miyazaki, replication)	Urbanschitz 2021 (34966859); Otaki 2021 (34803082)
Cobb / alignment	inter 0.88 / intra 0.83; **mixed-reader ∼0.55**	Sevin 2025 (41011045); Lee 2017 (28574884)
Cobb (radiograph borrow)	inter 0.886 / intra 0.958, SEM 1.8°, MDC 5.0°	Marques 2019 (31668048) *[borrow]*
Canal AP diameter	ICC 0.72–0.99	Rüegg 2015 (25525959); Griffith 2024 (38617159)
Canal-stenosis grade	ICC 0.72–0.80, κ 0.60–0.62	Kang 2011 (21700974)
Disc height (mid-disc)	ICC 0.91–0.95	van Santbrink 2026 (41567758)
Vertebral-body height	*no cervical-MRI ICC exists*; lumbar 0.97	Germann 2022 (36576545) *[borrow]*
Space available for cord (SAC, mm)	*no primary cervical ICC exists*	— (negative)

3 Why the baseline is insufficient, and why our approach is justified

- **It does not scale.** Minutes of specialist time per case on repetitive geometry, at volume.
- **It is not reproducible.** The same study yields materially different numbers between readers (Section 2); a threshold-based decision sitting on a measurement with ICC 0.35–0.66 inherits that noise.
- **Our (non-LLM) AI approach removes both costs by construction.** Automated segmentation (TotalSpineSeg / Spinal Cord Toolbox / SPINEPS) + deterministic geometric measurement gives *perfectly reproducible* numbers in seconds, and every flag is tied to a cited threshold (no black-box verdict). It is consistency + speed + auditability, not a diagnosis — outputs are “flagged for physician review.”

4 Honest provenance (the negatives are themselves the argument)

Several baseline figures are *borrows* (vertebral-body-height ICC is lumbar; some Cobb/disc figures are radiograph/CT), flagged above. More tellingly, for **vertebral-body height** and **SAC in millimetres** *no cervical-MRI inter-observer study exists at all.* These gaps cut in our favour: the manual baseline is so under-characterised that, for some of the exact measurements our pipeline produces deterministically, the literature cannot even state how much two humans would disagree. We automate what has not been reliably measured by hand.

References

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